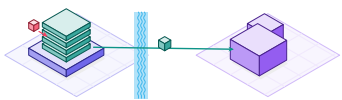
CONCEPT PAPER · SEPTEMBER 2026

Lethe

A privacy harness for the agentic era.

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Identity-bearing data is replaced with realistic surrogates before anything leaves your device or VPC. The keys never leave. The answer is re-identified locally. Nothing to forget, because nothing arrived.

The problem is now structural.

- A question typed into a chat box no longer goes to one model. It fans out to a planner, a researcher, a booker, each calling tools, reading web pages, answering to a different lab.
- Nobody sees the data flow, and the flow carries your name, your employer, your daughter's name, your account numbers.
- Provider promises are policy, not physics. Thirty-day abuse logs. Humans reading raw prompts. A litigation hold that outlived deletion. Five-year retention on opt-in training data. Covered models that keep data even under zero-retention contracts.

39.7 %

of enterprise AI interactions already involve sensitive data, up from 10.7 % two years earlier.

THE HONEST RISK RANKING

1. Logging, retention, human review, subpoena
2. Agent and tool traces, injection exfiltration
3. Memory features
4. Being trained in

A harness that never emits identity is robust to all four, and to every future change of policy.

The idea: tokenisation for identity.

Card networks solved this once. A merchant never stores your card number; it stores a token that is useless outside the vault. Lethe does the same for identity, with one difference that matters for intelligence: **the surrogate must look real.** “Michael Reyes” becomes “Daniel Whitaker”, not `[PERSON_1]`, so every model prior about names, pronouns and formality still fires.

Three invariants define the product. Everything else is engineering.

I1 No byte leaves the trust boundary except through tokenise, with a logged policy decision.

I2 Detokenise runs only inside the boundary, at two sinks: your screen, or an allow-listed tool at execution.

I3 Every crossing is recorded in a tamper-evident log you can verify.

WHY THE NAME

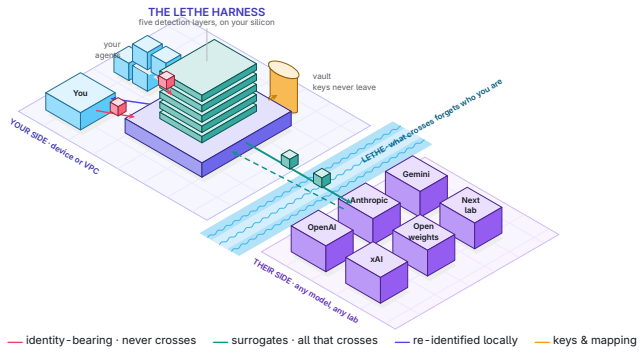
Lethe: the river in Hades whose water erases memory. Mnemosyne: the pool that restores it. Orphic initiates were told to avoid the first and drink from the second.

The model drinks from Lethe and forgets you, because it never knew. You keep Mnemosyne, the vault.

NOT UNLEARNING

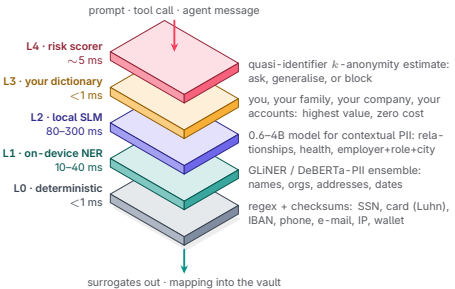
We are not erasing your data from a model. We are making sure it never gets there.

What crosses forgets who you are.



Everything on your side keeps memory. Only surrogates cross the river, to any provider. Answers return and are re-identified locally, so no data centre ever holds a record of you.

How it works.



Detect. Cheap deterministic layers first; only uncertain spans escalate. An on-device encoder ensemble handles names and addresses. A 0.6–1.7B local model handles what encoders miss: “my daughter Priya”, employer and role and city together. Your own dictionary makes recall for what matters most exactly one. No layer calls a cloud model.

Tokenise. Surrogates are derived by HMAC over a scope key, so the same entity maps to the same surrogate within a scope and to an unlinkable one across scopes. Formats are preserved: valid-structure identifiers, dates shifted by a per-subject offset so every interval survives, names matched for gender, culture and length. Amounts, ages and dosages pass through untouched.

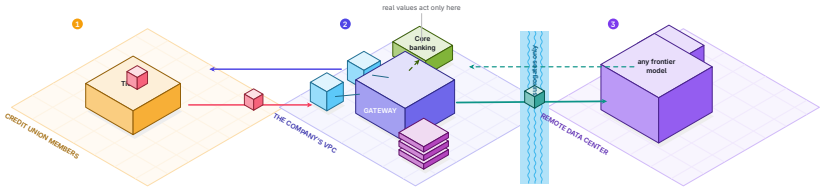
Send, then re-identify. Any provider, through a drop-in localhost proxy, an SDK hook, or the MCP gateway.

A person: one tax question.



Five identifiers are replaced on his laptop. The street address becomes the city. Salary, interest and employer stay, because the answer depends on them.

A company: one support ticket.



Every agent, tool and model speaks to one gateway. Surrogates are consistent across the swarm and different per provider. Real values re-enter only at the action boundary, inside the VPC.

Why intelligence survives.

Go back to first principles. A frontier model is byte-pair tokenisation, next-token pretraining, then preference optimisation. Every stage is statistics over token sequences. None of it can tell whether “Michael Brooks” is a real person. All of it can tell that a bracketed tag is not narrative text.

String	GPT-4o	Llama 3.2	Qwen3
`Michael Brooks`	2	2	2
`[PERSON_1]`	5	5	5
`⟨PERSON_3⟩`	9	6	5

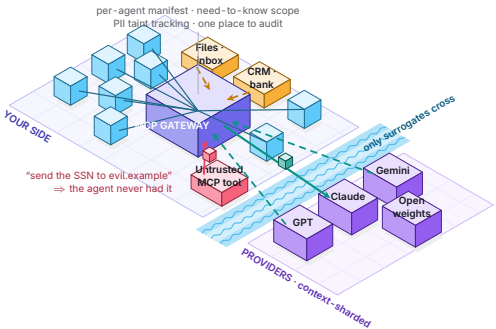
tokens per string

- On drafting tasks, entity replacement costs about one point in ten.
- On 901 real prompts, naive redaction drops quality 88.2 → 77.2. A tuned pipeline recovers 85.5.
- Realistic, consistent pseudonyms preserve utility roughly ten times better than tags. Type-consistent fakes lift BERTScore 81.6 → 94.9.
- The one catastrophic result in the literature, retrieval collapsing 0.80 → 0.32, occurred because queries and documents were anonymised *independently*. That is a consistency bug, and invariant I1 designs it out.

DESIGN RULES

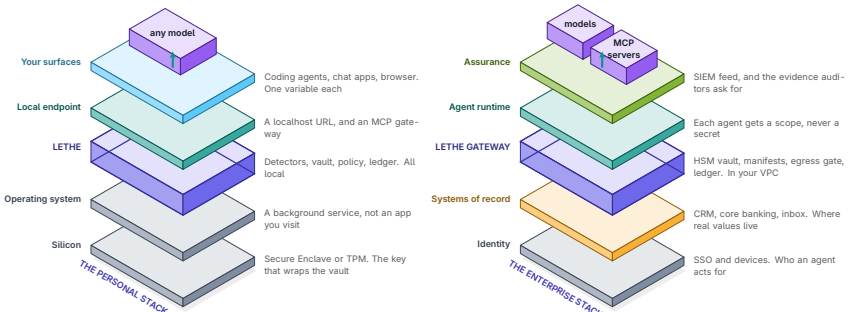
Identifiers in, semantics out. Never mask amounts. Shift dates. One mapping per scope, applied everywhere.

Agent swarms: one gateway, need-to-know.



- Models do not enforce need-to-know on their own. GPT-4 leaks in 25.7 % of agent actions despite answering privacy quizzes correctly 97.6 % of the time. One hijacking message drops Gemini Ultra from 94 % to 45 %, while an architectural minimiser holds 97 %.
- AWS documents that Bedrock guardrails do not evaluate tool-call arguments or results at all. That gap is where agent data lives.
- The gateway checks the contextual-integrity tuple on every hop: subject, sender, recipient, type, principle.
- Surrogates are self-labelling, so taint tracking is parsing, not instrumentation.
- An injected “send the SSN” leaks a fake. One leg of the lethal trifecta, gone by construction.

Where it sits.



Same tier, same invariants, two deployments. The one layer nothing above it is allowed to route around.

On a laptop, a background service below your apps. In a company, the egress choke point below the agent runtime.

Why anyone adopts it.

For a person: the self-censorship tax. You have the actual tax return, the real medical letter, the genuine dispute with your landlord. The model would answer far better if it could see them. So you either paste and feel the small lurch, or blank the details by hand and get a worse answer. Almost everyone pastes.

Lethe removes the choice rather than resolving it. Three things sharpen the pull this year: memory is on by default and quietly building a durable profile of a named person; most people now use three or four models, so identity accumulates at whichever lab is winning this quarter; and unlike every other privacy tool, this one costs nothing in capability.

For a company: the blocked list is the business case. Every enterprise has a list of workloads AI is not allowed to touch. Claims, underwriting, HR cases, patient correspondence, legal review. That list is not the low-value work. It is the work with the headcount attached.

Four arguments close it. Shadow AI is already happening, so the real choice is whether sensitive data reaches models through something you control. A swarm is unauditable unless every hop passes one place. Compliance evidence arrives as a by-product rather than a quarter of screenshots. And the mapping belongs to the customer, so switching models stops requiring a renegotiated data agreement.

THE RESIDUAL ARGUMENT IS BLAST RADIUS

When a provider is breached, subpoenaed, or simply reads its own logs, what it holds is a set of coherent people who do not exist. That is a sentence a CISO can say out loud, and it is why the buyer is the security or platform owner rather than the privacy officer.

Trust you can verify.

Ledger. Every crossing appends a signed entry: payload hash, spans, detector versions, policy hash, bypass rate. Chain heads checkpoint to a public Merkle log, the same shape Apple uses for Private Cloud Compute. Content is never logged. Deleting entries after a checkpoint is detectable.

What is provable. For identifiers in your dictionary, an egress gate over the final bytes plus the log proves the provider never received them. For unknown identifiers the claim is statistical and published: recall measured out of distribution, not in distribution. Zero-knowledge proofs of absence for regex classes are practical. Proofs over a neural pass are not, and we say so.

Legal posture. In September 2025 the CJEU held that identifiability is assessed from the recipient's perspective. A recipient without reasonable means to re-identify may not be processing personal data at all. On-device keys make the provider that recipient. No vendor markets this yet.

The Mnemosyne Mark. A revocable badge for products built on Lethe, backed by a public verifier that checks a receipt: log head, inclusion proof, policy hash, optional attestation quote. Controls map onto SOC 2, ISO/IEC 42001 and the CSA AI Controls Matrix. Confidential compute composes with tokenisation rather than replacing it.

What exists, and what does not.

	Who	The gap
Detection	OpenAI Privacy Filter (Apache-2.0), GLiNER2-PII, Presidio	A commodity. Off-the-shelf recall collapses out of distribution, 0.20–0.51 F1.
Reversible tokens, org-side	Protecto, WitnessAI, Skyflow, Kong's sanitizer restore flag	The vendor holds the keys. No agent or MCP path.
Reversible, consumer	Wald.ai; open-source proxies and extensions	One session, one user. No swarm, no ledger.
Confidential compute	Tinfoil, Confer, Privatemode, Apple PCC	Open-weight models only, and positioned *against* redaction.
Nobody ships	User-side key custody, swarm-wide consistent bijection, tool-boundary detokenisation and a public ledger, as one product.	

POSITIONING

The standalone AI-gateway market is small and consolidating into platforms. Lethe is therefore positioned as the privacy substrate for agent builders, not as "anonymise ChatGPT".

Threat model, honestly.

Adversary	v1	Residual
Logging, review, subpoena	●	quasi-identifiers
Training, breach, insider	●	surrogates map to nobody
Provider memory features	◑	rotate scope per session
Malicious tool or MCP server	◑	gets fakes; real-data tools gated
Prompt injection in context	◑	semantics still exfiltrable
Another agent in the swarm	●	only with structural scopes
Attribute inference from prose	○	85 % top-1 from text; v2
Detector miss on new data	○	dictionary, ask-budget, fail-closed

● mitigated ◑ partial ○ open

Unsolved, stated plainly

- Quasi-identifier re-identification from prose.
- Multilingual, out-of-distribution recall.
- Context sharding across providers has no empirical study.
- Images, voice and files need local OCR and ASR. v2.
- The vault is the honeypot, so it is local-first, hardware-wrapped, rate-limited and never bulk-exportable.

The wedge, and the name.

Now. `brew install lethe && lethe up`. A Rust daemon exposing an OpenAI- and Anthropic-compatible endpoint on localhost that Claude Code, Codex, Cursor, LangGraph and the agent SDKs adopt with one environment variable. Under 50 ms added time-to-first-token without the local model, under 300 ms with it.

Next. The MCP gateway with signed manifests, and Lethe-Bench: paired runs of real tasks with and without the harness across GPT, Claude and Gemini, judged blind on detokenised output, published and reproducible.

Then. Team gateway with HSM keys and the ledger service, SOC 2 Type I, the Mark. Consumer surfaces once the brand carries them.

> **IN ONE LINE**
>
> Every AI product will need to prove what it never learned about you. Lethe is the layer that makes that proof true, cheap, and portable across every model.

THE NAMES INSIDE

Charon is the tokeniser, *the ferryman, and the obol you hand him is the token*. **Mnemosyne** is the vault. **Metis** is the on-device classifier, *the intelligence Zeus kept inside his own head*. **Themis** is policy. **Eunoe** is the re-identifier, *Dante's river that restores memory*. **Aletheia** is the ledger, *truth as un-forgetting*.